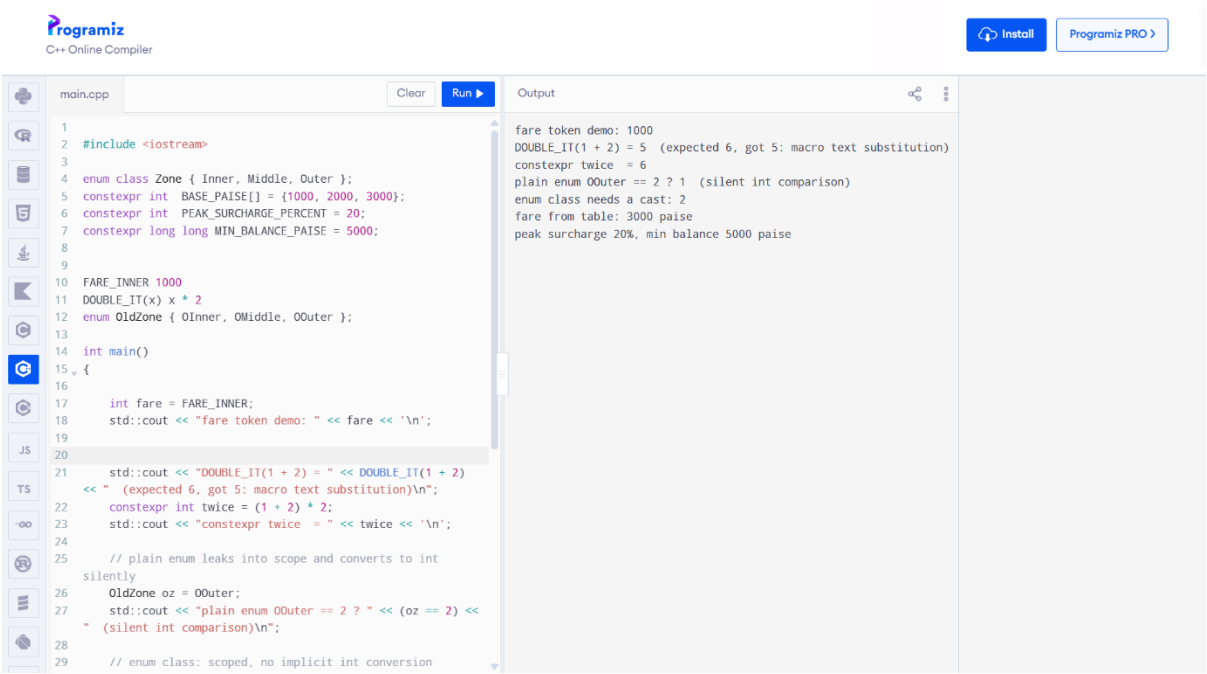
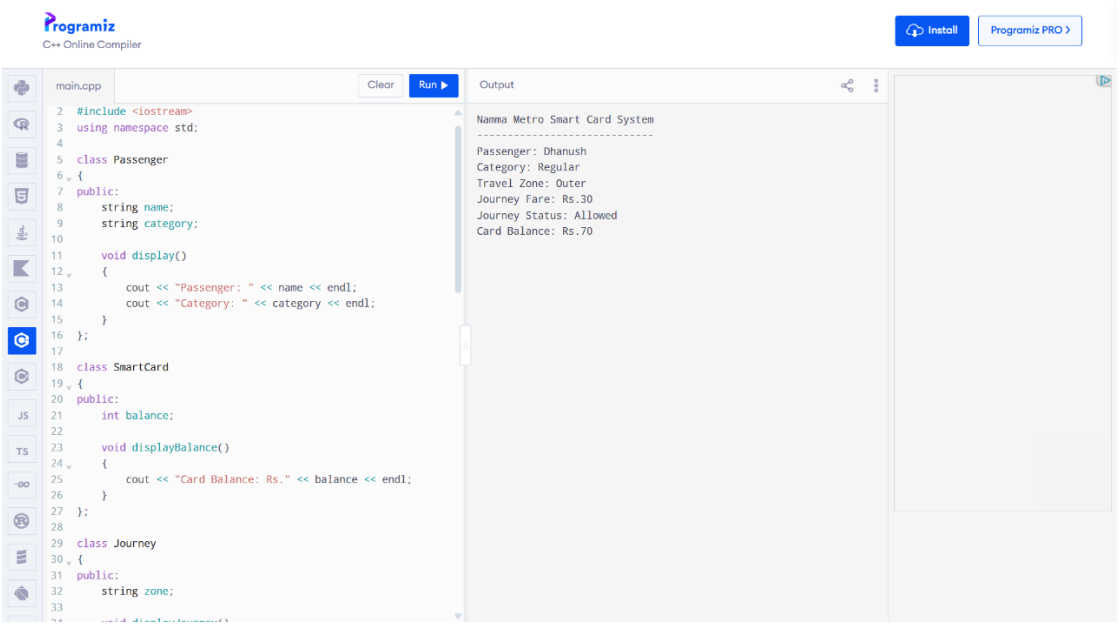
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C01 ASSIGNMENT

TASK 1

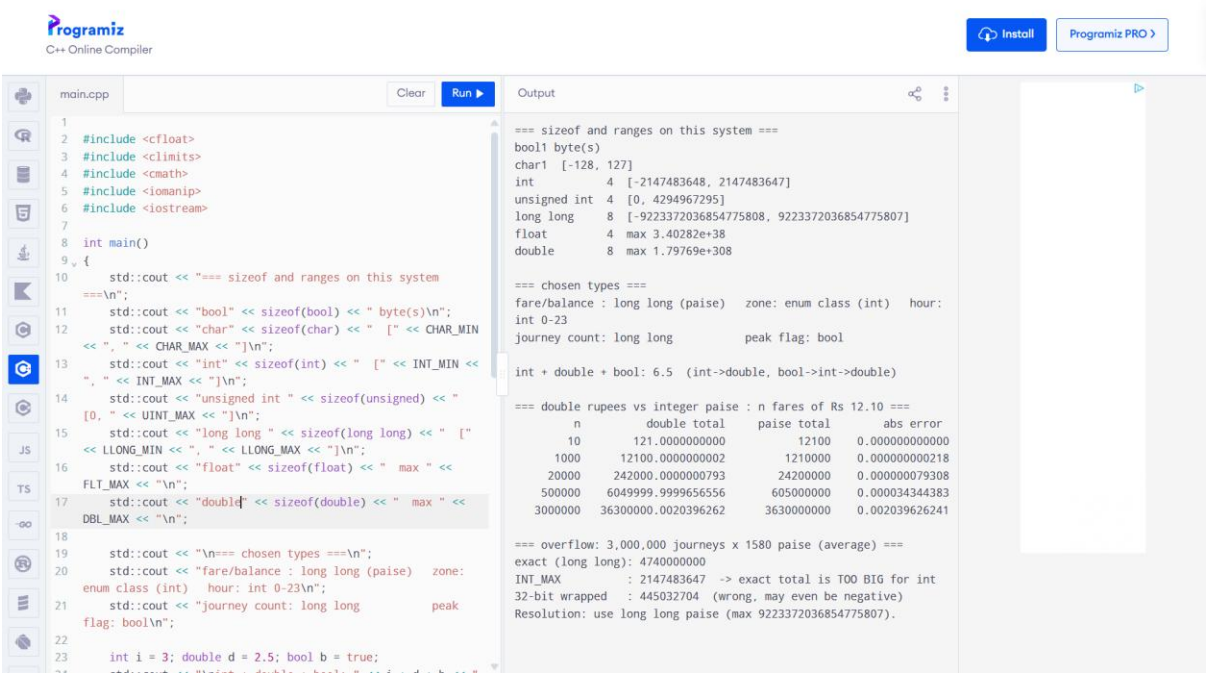


TASK 2



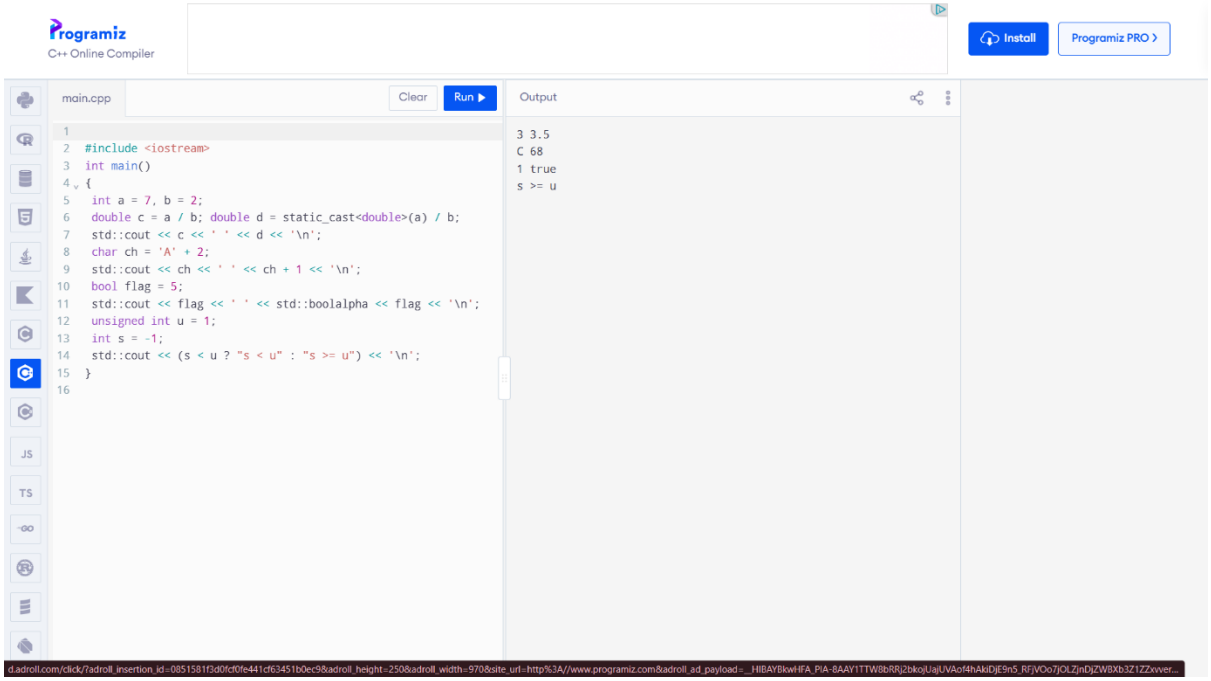
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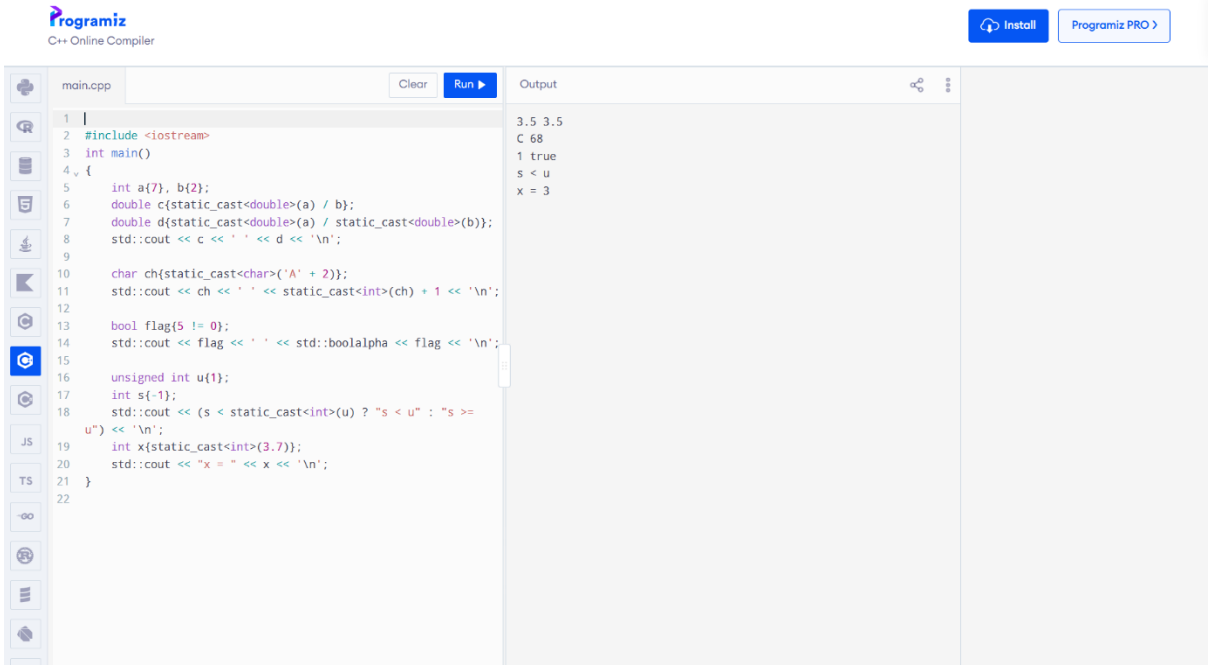
TASK 3



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TASK 4





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TASK 5

Programiz
C++ Online Compiler

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main.cpp | Clear | Run

```cpp
#include <iostream>
int main()
{
 int x = 5, y = 3;
 std::cout << x + y * 2 << ' ' << (x + y) * 2 << '\n';
 int z = x++ + ++y;
 std::cout << z << ' ' << x << ' ' << y << '\n';
 std::cout << (x > y ? x : y) << ' ' << x % 4 * 2 << '\n';
 std::cout << 10 / 3 << ' ' << 10 % 3 << ' ' << -10 / 3 << ' ' << -10 % 3 << '\n';
 int k = 0;
 for (int i = 0; i < 10; i += 3) k += i;
 std::cout << k << '\n';
}
```

Output

```
11 16
9 6 4
6 4
3 1 -3 -1
18
```

Programiz
C++ Online Compiler

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main.cpp | Clear | Run

```cpp
#include <iostream>

bool isPeakReadable(int hour) { return (hour >= 8 && hour < 10) || (hour >= 17 && hour < 20); }

bool isPeakNoBrackets(int hour) { return (hour >= 8 && hour < 10) || (hour >= 17 && hour < 20); }

constexpr bool PEAK[24] =
{false,false,false,false,false,false,false,false,
                          true, true,
false,false,false,false,false,false,false,
                          true, true, true,
                          false,false,false,false};

bool isPeakTable(int hour) { return PEAK[hour]; }

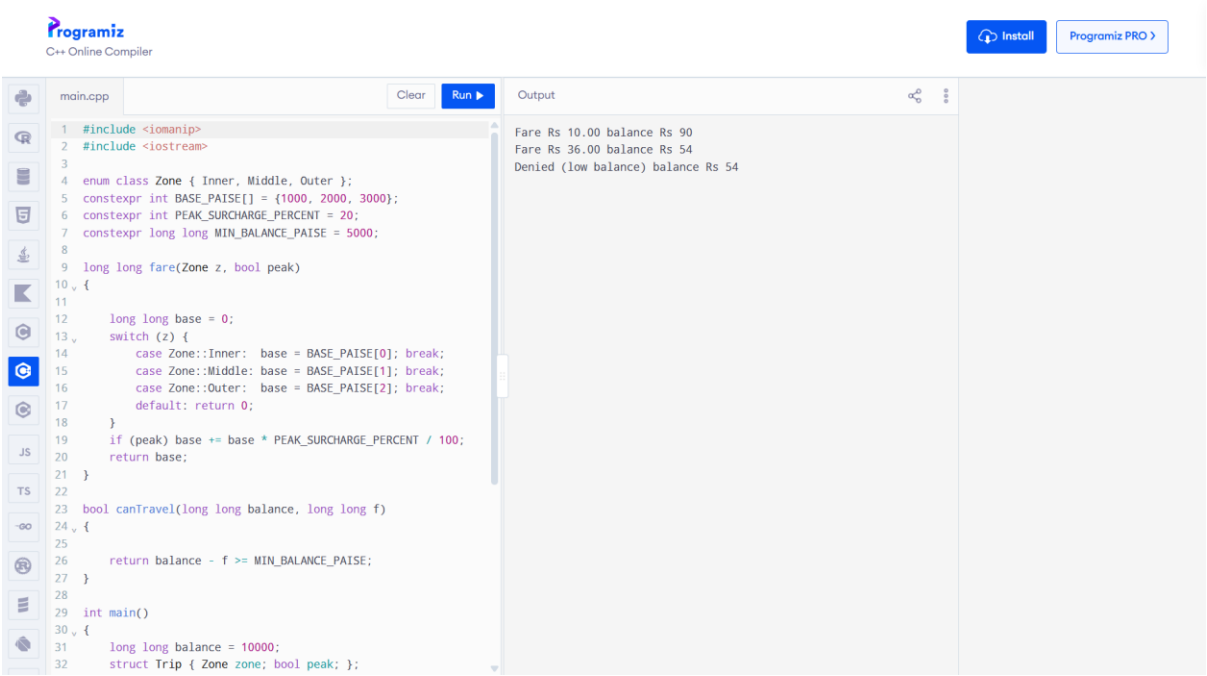
int main()
{
    int checks[] = {7, 8, 9, 10, 16, 17, 19, 20, 23};
    std::cout << "hour  readable  table\n";
    bool allSame = true;
    for (int h : checks) {
        std::cout << "  " << h << (h < 10 ? " " : "") << "
    " << isPeakReadable(h) << "  " << isPeakTable(h) << '\n';
        if (isPeakReadable(h) != isPeakTable(h) ||
isPeakReadable(h) != isPeakNoBrackets(h)) allSame = false;
    }
    std::cout << "All versions agree: " << (allSame ? "yes"
```

Output

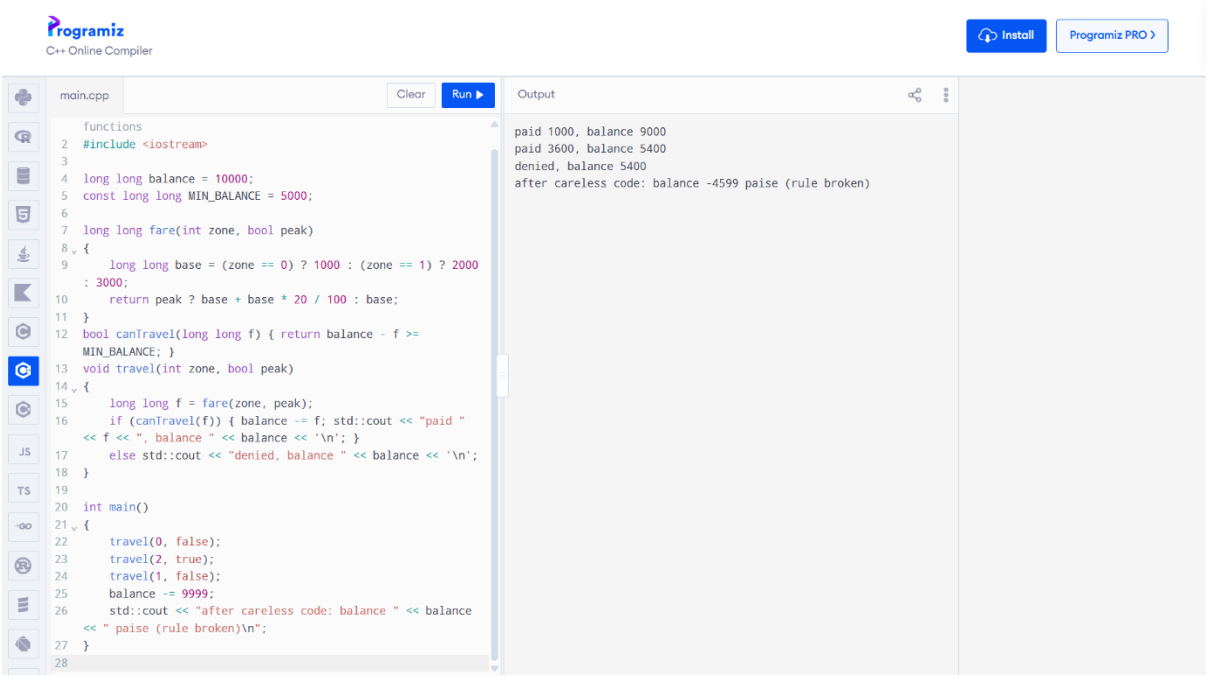
```
hour  readable  table
  7       0       0
  8       1       1
  9       1       1
 10       0       0
 16       0       0
 17       1       1
 19       1       1
 20       0       0
 23       0       0
All versions agree: yes
Outer, 17:00, student fare = 1800 paise
a=5 b=7 i=7
```

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TASK 6

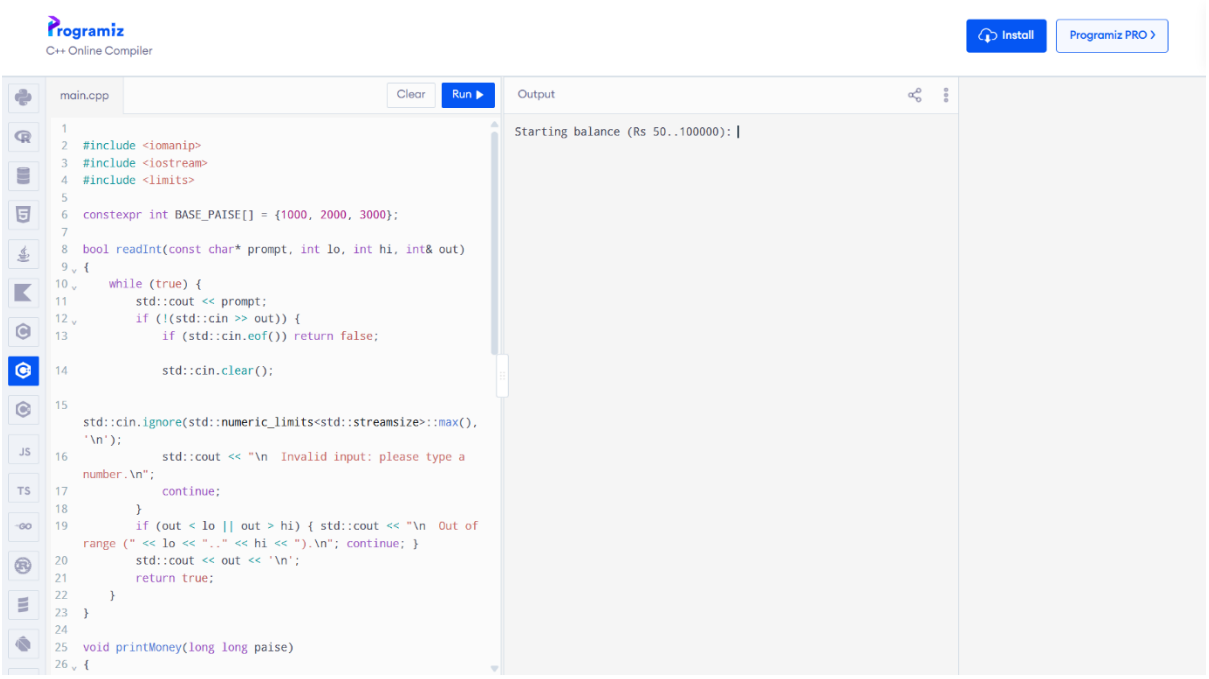


TASK 7



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TASK 8



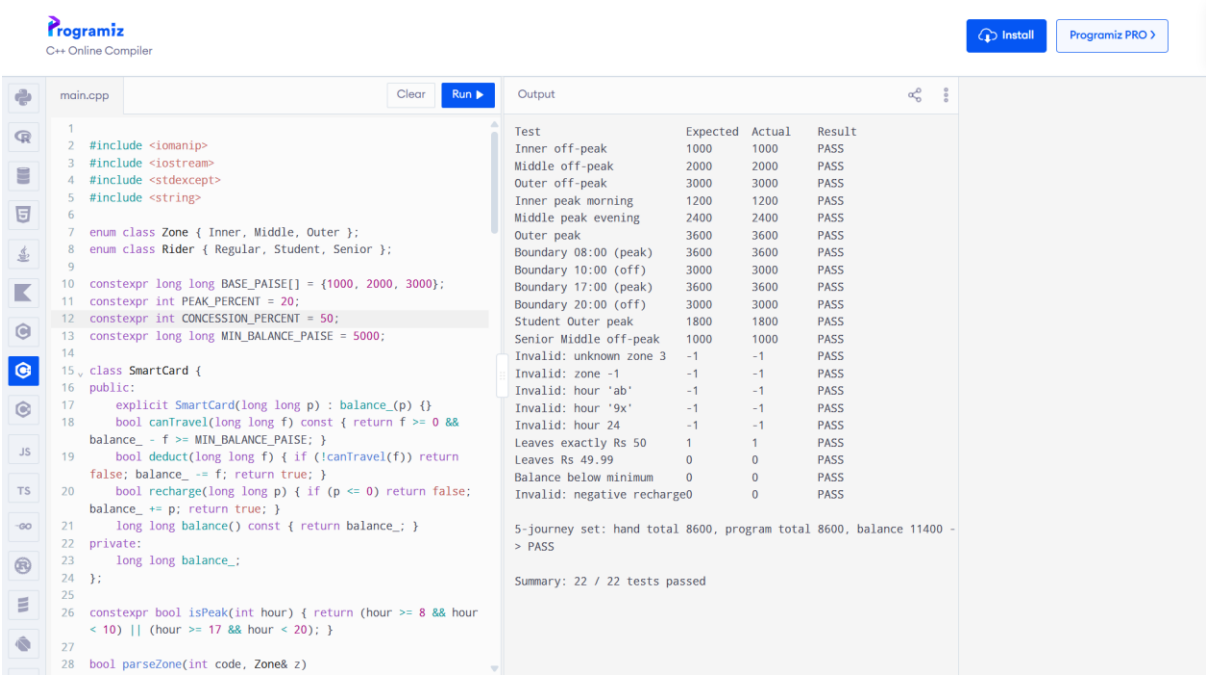

```cpp
#include <iomanip>
#include <iostream>
#include <limits>

constexpr int BASE_PAISE[] = {1000, 2000, 3000};

bool readInt(const char* prompt, int lo, int hi, int& out)
{
    while (true) {
        std::cout << prompt;
        if (!(std::cin >> out)) {
            if (std::cin.eof()) return false;
            std::cin.clear();
            std::cin.ignore(std::numeric_limits<std::streamsize>::max(), '\n');
            std::cout << "\n  Invalid input: please type a number.\n";
            continue;
        }
        if (out < lo || out > hi) { std::cout << "\n  Out of range (" << lo << ".." << hi << ").\n"; continue; }
        std::cout << out << '\n';
        return true;
    }
}

void printMoney(long long paise)
{
```

TASK 9




```cpp
#include <iomanip>
#include <iostream>
#include <stdexcept>
#include <string>

enum class Zone { Inner, Middle, Outer };
enum class Rider { Regular, Student, Senior };

constexpr long long BASE_PAISE[] = {1000, 2000, 3000};
constexpr int PEAK_PERCENT = 20;
constexpr int CONCESSION_PERCENT = 50;
constexpr long long MIN_BALANCE_PAISE = 5000;

class SmartCard {
public:
    explicit SmartCard(long long p) : balance_(p) {}
    bool canTravel(long long f) const { return f >= 0 && balance_ - f >= MIN_BALANCE_PAISE; }
    bool deduct(long long f) { if (!canTravel(f)) return false; balance_ -= f; return true; }
    bool recharge(long long p) { if (p <= 0) return false; balance_ += p; return true; }
    long long balance() const { return balance_; }
private:
    long long balance_;
};

constexpr bool isPeak(int hour) { return (hour >= 8 && hour < 10) || (hour >= 17 && hour < 20); }

bool parseZone(int code, Zone& z)
```

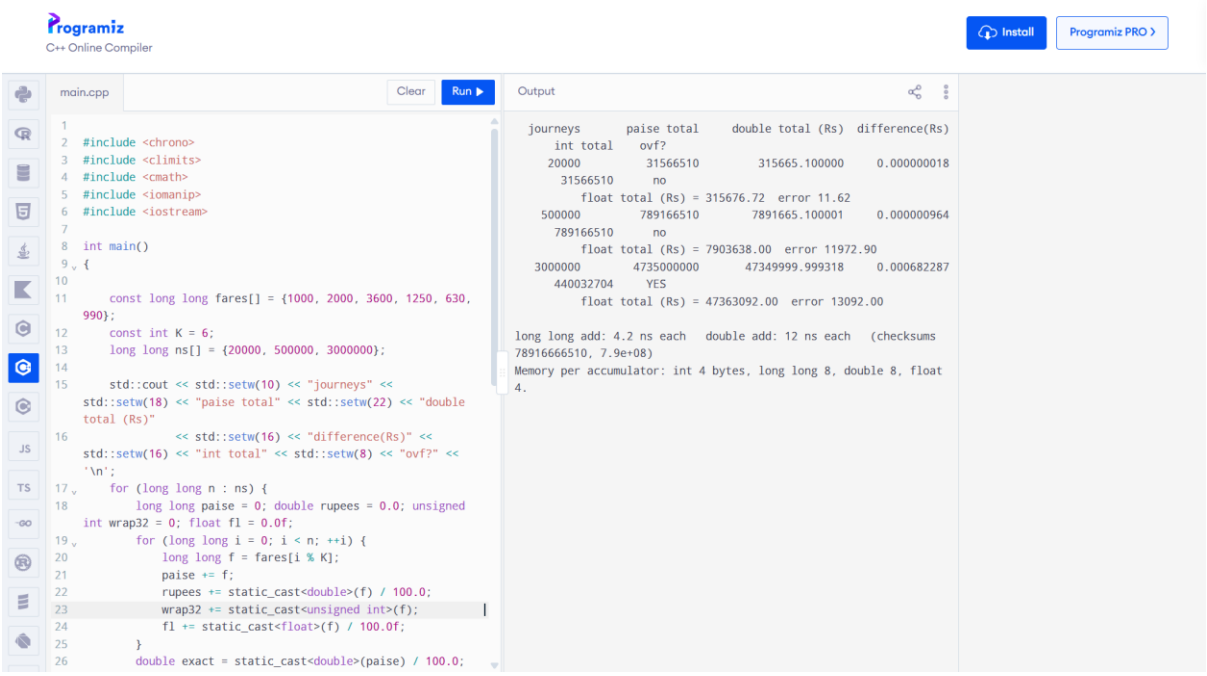
```
Test                        Expected  Actual    Result
Inner off-peak              1000      1000      PASS
Middle off-peak             2000      2000      PASS
Outer off-peak              3000      3000      PASS
Inner peak morning          1200      1200      PASS
Middle peak evening         2400      2400      PASS
Outer peak                  3600      3600      PASS
Boundary 08:00 (peak)       3600      3600      PASS
Boundary 10:00 (off)        3000      3000      PASS
Boundary 17:00 (peak)       3600      3600      PASS
Boundary 20:00 (off)        3000      3000      PASS
Student Outer peak          1800      1800      PASS
Senior Middle off-peak      1000      1000      PASS
Invalid: unknown zone 3     -1        -1        PASS
Invalid: zone -1            -1        -1        PASS
Invalid: hour 'ab'          -1        -1        PASS
Invalid: hour '9x'          -1        -1        PASS
Invalid: hour 24            -1        -1        PASS
Leaves exactly Rs 50        1         1         PASS
Leaves Rs 49.99             0         0         PASS
Balance below minimum       0         0         PASS
Invalid: negative recharge0           0         PASS

5-journey set: hand total 8600, program total 8600, balance 11400 -> PASS

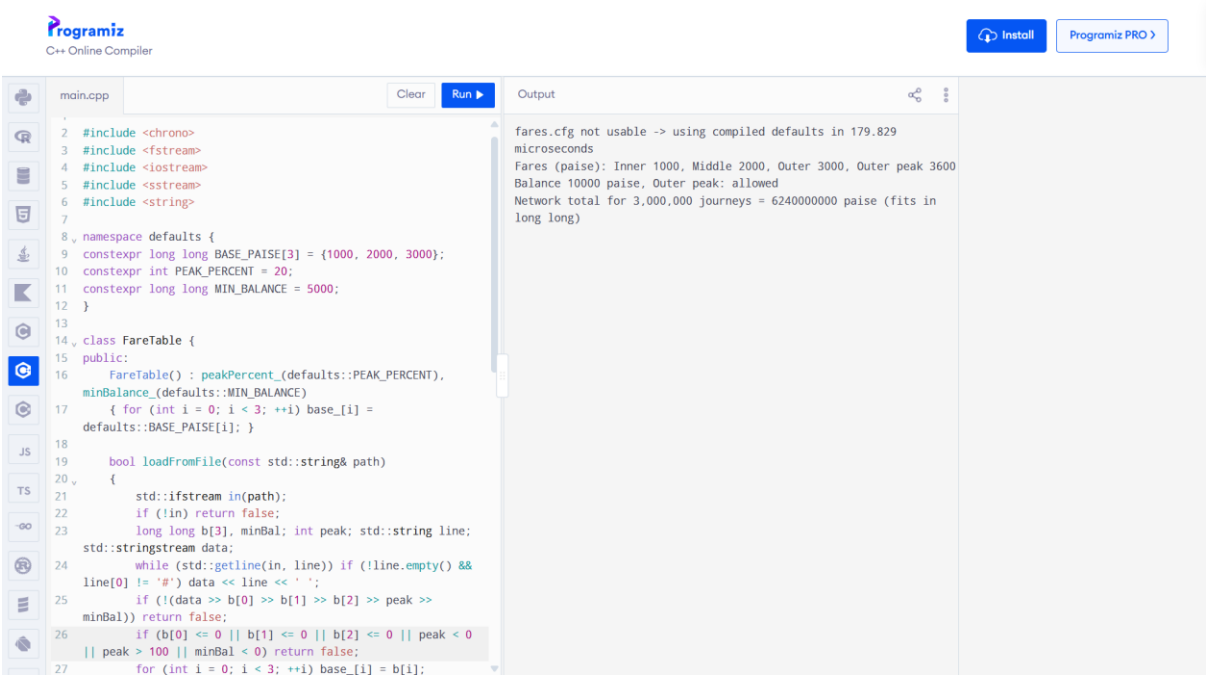
Summary: 22 / 22 tests passed
```

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TASK 10



TASK 11



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TASK 12

